

Research paper

Docking assistance method for autonomous berthing by backward-time imitation learning and kernel density estimation based on AIS data

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ABSTRACT

Autonomous berthing is desirable to assist or replace human operators to reduce the burden on navigation officers and resolve labor shortages. Over the past few decades, while researchers have proposed various methods to realize autonomous berthing, no online trajectory-planning method exists that reflects the intuition of skilled captains and is guaranteed to be constrained to terminal states. Additionally, most previous studies assumed a single reference trajectory, which tended to lack important information on the permissible deviation from the reference trajectory. This study presents a docking assistance method that integrates backward-time imitation learning (BTIL) and kernel density estimation (KDE). BTIL enabled a computational agent to generate captain-like docking trajectories without tuning the objective functions while imposing implicit constraints on terminal states. The proposed method offered trajectory and guidance distributions by applying multidimensional KDE to the BTIL results. The trajectory distributions indicated the area through which the ship should pass for successful berthing, whereas the guidance distributions indicated the velocities the ship should maintain at arbitrary locations. The performance of the docking assistance method was verified using Automatic Identification System (AIS) data from Shinmoji Port in Fukuoka Prefecture, Japan.

Nomenclature

s, s'	State
S	Set of states
a	Action
A	Set of actions
$r(s)$	Reward function
$p(s' s, a)$	Transition probability to a state s' , from a state s taking an action a
π	Policy (decision-making rule)
π^*	Optimal policy
$\pi(a s)$	Probability of taking an action a in a state s under a stochastic policy π
t	Discrete time step
$\bar{t}$	Reversed discrete time step in a backward-time simulation
s_t	State at time t
a_t	Action at time t
r_{t+1}	Reward at time $t+1$, typically due to s_t and a_t
R_t	Return (summation of discounted rewards) following time t
γ	Discount rate
T	Time horizon (step size of an episode)
θ	Weight parameter of an actor network
ϕ	Weight parameter of a critic network
ω	Weight parameter of a discriminator network

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$\pi_\theta(a s)$	Policy under the parameter θ
$V_\phi(s)$	Value under the parameter ϕ
$\mathcal{D}_\omega(s)$	Output of a discriminator under the parameter ω
$\mathcal{L}^A(\theta)$	Loss function of an actor
$\mathcal{L}^C(\phi)$	Loss function of a critic
$\mathcal{L}^D(\omega)$	Loss function of discriminator
$\mathcal{L}^{\text{clip}}(\theta)$	Surrogate objective of PPO
$\mathcal{L}^{\text{ent}}(\theta)$	Entropy bonus
ϵ	Clipping parameter of PPO
c_e	Entropy coefficient of PPO
$\mathcal{I}$	Invariant set in TRAIL
$T_{\mathcal{I}}$	Interval of invariant set in TRAIL
λ	Tuning parameter of TRAIL
ρ	Dropout rate
s_E	State of an expert
π_E	Policy of an expert
N_{env}	Number of parallel environments
N_{smpl}	Number of sample trajectories
$\zeta_s^{(i)}$	i -th trajectory of an agent
$\zeta_E^{(i)}$	i -th trajectory of an expert

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$O - XY$	Global coordinate system
X	Latitudinal position of the ship
Y	Longitudinal position of the ship
Ψ	Heading angle of the ship
$o - xy$	The own-ship-fixed coordinate system
S_{base}	Set of states corresponding to the ship and destination
S_{detect}	Set of states corresponding to detection of obstacles
d_x	Longitudinal distance from the ship to the destination
d_y	Transverse distance from the ship to the destination
$d_{ }$	Component parallel to the terminal berth of the distance from the ship to the destination
$d_{\perp}$	Component vertical to the terminal berth of the distance from the ship to the destination
v_w	Wind speed
θ_w	Wind direction
w_x	Wind velocity in surge direction
w_y	Wind velocity in sway direction
v_c	Current speed
θ_c	Current direction
c_x	Current velocity in surge direction
c_y	Current velocity in sway direction
d_i	Distance from the own ship to an obstacle on the detection line i
d_{lim}	Length of the detection lines
$\Delta\varphi$	Interval of the detection lines
ψ	Heading angle of the own ship relative to the terminal berth
u	Surge velocity of the own ship
v	Sway velocity of the own ship
r	Angular velocity (yaw rate) of the own ship
$\dot{u}$	Surge acceleration of the own ship
$\dot{v}$	Sway acceleration of the own ship
$\dot{r}$	Angular acceleration of the own ship
$ \dot{u} _{max}$	Maximum value of surge acceleration
$ \dot{v} _{max}$	Maximum value of sway acceleration
$ \dot{r} _{max}$	Maximum value of angular acceleration
$\mathcal{B}(x; \alpha, \beta)$	Beta distribution of the first kind
α^j	Parameter α corresponding to j -th action
β^j	Parameter β corresponding to j -th action
N	Total time steps of the generated trajectories
h	Bandwidth of KDE
$[z]_E^{(i)}$	Time history of an arbitrary variable z in the i -th expert trajectory
$[\bar{z}]_E^{(i)}$	Reversed time history of an arbitrary variable z in the i -th expert trajectory
$f_{2D}(X, Y)$	Trajectory distribution calculated with 2D-KDE
$f_{3D}(X, Y, \Psi)$	Trajectory distribution calculated with 3D-KDE
$f_{4D}(X, Y, \Psi, z)$	Guidance distribution calculated with 4D-KDE

1. Introduction

Berthing, one of the most challenging tasks in maritime navigation, involves collision risks against quays and other ships and thus requires sophisticated and reliable maneuvers. During docking, disturbances, such as wind and currents, affect ships more severely than during usual navigation. Fluid dynamics become much more complex, especially at low speeds, making it difficult to predict the ship's maneuvering motion difficult. Although this difficult task is accomplished by experienced captains, a shortage of seafarers and an aging workforce have highlighted the need for autonomous berthing systems to assist or replace human operators.

Neural networks (NNs) are powerful tools that are expected to realize autonomous berthing. At the beginning, Yamato et al. (1990) proposed a docking control method based on an NN. Im and Nguyen (2018) enhanced its versatility; their NN-controller can be applied to different ports without retraining. Qiang et al. (2019) addressed unknown dynamics and external disturbances by introducing dynamic deep-rooted information to the NN controller. Shuai et al. (2019) proposed generating reliable docking data and feature selection to achieve automatic docking under various environmental disturbances. With the further development of NNs, deep reinforcement learning (DRL) has also been studied in recent years. NNs are generally trained using teaching data prepared in advance, whereas DRL improves the action strategy through trial and error without teaching data. Shimizu et al. (2022) proposed

combining trust region policy optimization with behavioral cloning and demonstrated an online docking method that considered port geometry and wind disturbance. Higo et al. (2023) presented online path-tracking control using a double deep Q-network and demonstrated that a large car ferry safely berthed/unberthed under wind disturbance at real-ship experiments. However, such DRL-based methods require the design of reward functions, and it can be challenging to find appropriate rewards for reproducing sophisticated maneuvers by skilled captains, as reported by Higaki and Hashimoto (2023). To resolve this problem, they introduced imitation learning (IL) instead of DRL to generate captain-like route plans for collision avoidance. However, this method cannot impose constraints on the terminal states, which is a problem in berthing situations because the collision risk gets the highest in the terminal states (i.e., the docking phase). In addition, the black-box characteristic is another issue in NN-based approaches.

Solving docking tasks as optimal control problems (OCP) is common practice. Shouji et al. (1992) were the first to formulate docking as a minimum-time problem and proposed deriving the optimal solution using a sequential conjugate gradient restoration algorithm. Maki et al. (2020) established an offline control method that minimized the required time while considering collision risks using a covariance matrix adaptation evolution strategy (CMA-ES). Following this research, Miyauchi et al. (2022) enabled docking at ports with complicated geometries while considering collision risks and wind disturbances. Suyama et al. (2022) proposed a human-like trajectory planning method by introducing checkpoint conditions based on maneuvering data at real sea. However, these methods require extensive time (ranging from hours to days) to calculate an optimal route, which precludes real-time planning. Consequently, once a ship fails to follow the optimal route, returning to the original route may be difficult.

Model predictive control (MPC) has been increasingly used to accelerate solving OCPs. Mizuno et al. (2015) proposed a quasi-real-time optimal control method that combined a multiple shooting algorithm with MPC. Martinsen et al. (2019) introduced a real-time MPC-based algorithm that considered spatial constraints. Li et al. (2020) enabled controlling the propeller revolution and rudder angle as inputs using nonlinear MPC. However, the performance of MPC depends on model accuracy, which requires precise modeling of each individual vessel. Moreover, these direct control approaches can render their intentions unclear to human operators.

The intentions of autonomous systems can be better understood by humans by dividing the docking tasks into trajectory planning and tracking control. Liao et al. (2019) proposed an improved artificial potential field method for trajectory planning and an adaptive fuzzy PID for tracking control, with both verified through simulation and field experiments. Bitar et al. (2020) established an automatic docking method for small-surface vehicles by interconnecting an optimization-based trajectory planner with a dynamic positioning tracking controller. Han et al. (2021) presented an extended dynamic window approach (EDWA) to perform motion planning that took ship dynamics and collision avoidance into account. Consequently, Han et al. (2022) showed that a potential field-based EDWA enabled real-time trajectory planning considering COLREGs. Wang et al. (2022) realized appropriate piloting by predicting a trajectory with a long short-term memory and tracking it using nonlinear MPC. Sawada et al. (2023) developed an automatic berthing system that consisted of path planning and a feedforward controller and demonstrated its performance under wind disturbance in a full-scale experiment at sea. Notably, all the above studies assumed the tracking of a single reference trajectory. However, an appropriate trajectory for docking is not necessarily unique, and offering a range of trajectories could be more intuitive for human operators. In addition, if the ship deviates significantly from the planned route, the autonomous system should switch to a fallback option (for example, switching to human operation). However, determining the extent of deviation tolerance using traditional methods is difficult.

Therefore, we propose a novel docking assistance method that

integrates backward-time imitation learning (BTIL) and kernel density estimation (KDE). BTIL essentially involves IL in backward-time simulations using Automatic Identification System (AIS) data. This approach enabled a computational agent to generate captain-like docking trajectories without tuning rewards or objective functions. In addition, by learning from reversed docking trajectories by going back in time from the destination, we imposed implicit constraints on terminal states, which has been a challenge in NN-based approaches. Another key aspect of this study is the probabilistic generation of numerous trajectories using a BTIL-based trajectory planner. This process enhanced the generalization performance of the proposed method, even with limited expert data. By applying two- and three-dimensional KDE (2D-KDE and 3D-KDE) to the generated routes, a trajectory plan was obtained as a distribution instead of a single route. The trajectory distributions implied the desirability of the docking approach, allowing for real-time evaluation of docking maneuvers. Furthermore, by adopting four-dimensional KDE (4D-KDE) to the generated trajectories, the proposed method can suggest appropriate ship motions, such as velocity and yaw rate—considered useful as both a support tool for human operators and an objective function for control systems.

The remainder of the paper presents the algorithm of BTIL with its basic theories: reinforcement learning (RL) and IL in Section 2. Section 3 provides an overview of the docking assistance method and implementation of BTIL and KDE. Section 4 presents the verification results using AIS data: a comparative study of IL and BTIL, demonstration of the trajectory distributions using 2D and 3D-KDE, and validation of the guidance distributions using 4D-KDE. Section 5 discusses the potential of the proposed method to treat external disturbances in the real world. Finally, the conclusions are presented in Section 6.

2. Theories

2.1. Reinforcement learning (RL)

RL generally assumes a Markov decision process (MDP; Bellman, 1957), which consists of a set of states S , a set of actions A , a transition probability $p(s'|s, a)$, and a reward function $r(s)$. If an agent at a state $s \in S$ takes an action $a \in A$, the next state s' is given independently on previous states or actions. RL aims to find the best policy that can maximize the summation of the rewards, defined as

$$R_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \quad (1)$$

Here, $\gamma \in (0, 1)$ denotes the discount rate, which implies that it is desirable for the agent to receive positive rewards as early as possible. Reward settings are important; however, setting appropriate rewards can be difficult, particularly when the task is complicated.

2.2. Imitation learning (IL)

IL is an effective method for optimizing policies without defining rewards. Task-relevant adversarial imitation learning (TRAIL; Zolna et al., 2020), an IL method, consists of discriminator and generator networks: the former discriminates whether a given trajectory is made by an expert, and the latter generates a policy that maximizes the summation of given rewards.

A discriminator parameterized by ω can be optimized by minimizing the binary cross entropy (BCE) as

$$\mathcal{L}^D(\omega) = BCE - \lambda \widehat{BCE} \mathbf{1}_{acc \geq \frac{1}{2}} \quad (2)$$

where $BCE = -E_{\pi}[\log(1 - \mathcal{D}_{\omega}(s))] - E_{\pi_E}[\log \mathcal{D}_{\omega}(s)]$,

$$\widehat{BCE} = -E_{\pi}[\log(1 - \mathcal{D}_{\omega}(\hat{s}))] - E_{\pi_E}[\log \mathcal{D}_{\omega}(\hat{s})],$$

$$acc(s, s_E) = \frac{1}{2T_{\mathcal{J}}} \sum_{t=0}^{T_{\mathcal{J}}-1} \left[\mathbf{1}_{\mathcal{D}_{\omega}(s_t) \geq \frac{1}{2}} + \mathbf{1}_{\mathcal{D}_{\omega}(s_{E,t}) < \frac{1}{2}} \right]$$

where π and π_E represent the policy of an agent and expert, respectively; $\mathcal{D}_{\omega}(s)$ denotes the probability that the state is made by the expert. In TRAIL, it is suggested to regard the states during the initial $T_{\mathcal{J}}$ steps ($\mathcal{J} = (s_0, s_1, \dots, s_{T_{\mathcal{J}}})$) as invariant to the task and exclude them from the loss function.

In contrast, a generator learns an optimal policy within the framework of DRL. Kostrikov et al. (2018) recommended using proximal policy optimization (PPO; Schulman et al., 2017) as the generator. As PPO generally takes an actor-critic style for continuous action spaces, the objective functions of the actor parameterized by θ and critic parameterized by ϕ are given as follows:

$$\mathcal{L}^A(\theta) = \mathcal{L}^{clip}(\theta) + c_e \mathcal{L}^{ent}(\theta) \quad (3)$$

$$\mathcal{L}^C(\phi) = MSE = E_{s \sim p, a \sim \pi_{\theta}} \left[\frac{1}{2} |V_{\phi}(s) - (r(s, a) + \gamma V_{\phi}(s'))|^2 \right] \quad (4)$$

Here, the actor loss is composed of the clipped gain $\mathcal{L}^{clip}(\theta)$ and the entropy bonus $\mathcal{L}^{ent}(\theta)$ multiplied by the entropy coefficient c_e ; the critic loss is calculated as the mean squared error (MSE) between the estimated value $V_{\phi}(s)$ and the target value. In general, IL cannot impose constraint conditions, unlike other numerical methods such as CMA-ES or MPC, because a generator simply aims to maximize the summation of rewards given by a discriminator. If an additional reward was defined as with DRL, such as a significant negative reward for the collision with the quay, we could force the agent to prevent such a collision. However, this technique would be unfavorable because it requires the tuning of reward functions and thus diminishes the strength of the IL.

2.3. Backward-time imitation learning (BTIL)

To address the issue of IL mentioned above, this study presents BTIL, which trains an agent with IL in backward-time simulations. Algorithm 1 shows the computational flow of BTIL, essentially the same as that of a general IL. One of the differences is that BTIL reverses the time histories of expert data. In this study, the reversed discrete time step is denoted as $\bar{t} = T - t$, where T is the time horizon of each expert trajectory. Another difference is that the agent always starts training from the terminal state of expert data; this approach is expected to enable constraints in the terminal state, which may be difficult in a general DRL or IL.

The effectiveness of backward-time machine learning has been suggested in prior studies, such as reverse value iteration (RVI; Zang et al., 2007), forward-backward reinforcement learning (FBRL; Edwards et al., 2018), and backward imitation and forward reinforcement learning (BIFRL; Pan and Lin, 2022). RVI was proposed to reduce the large memory and computational costs of traditional dynamic programming methods, such as value iteration; however, it was only applicable to simple problems with small and discrete states. By contrast, BTIL can be applied to complicated problems involving continuous states and actions within an IL framework. Both FBRL and BIFRL introduced an NN to estimate the backward dynamics model, $s_t \leftarrow f(s_{t+1}, a_{t+1})$. In contrast, BTIL targets problems in which the forward/backward dynamics are known. In other words, it cannot work if the state transition model is unknown but is easily implementable in trajectory planning, where ship velocities are computed from accelerations and positions are computed from velocities using the Euler method.

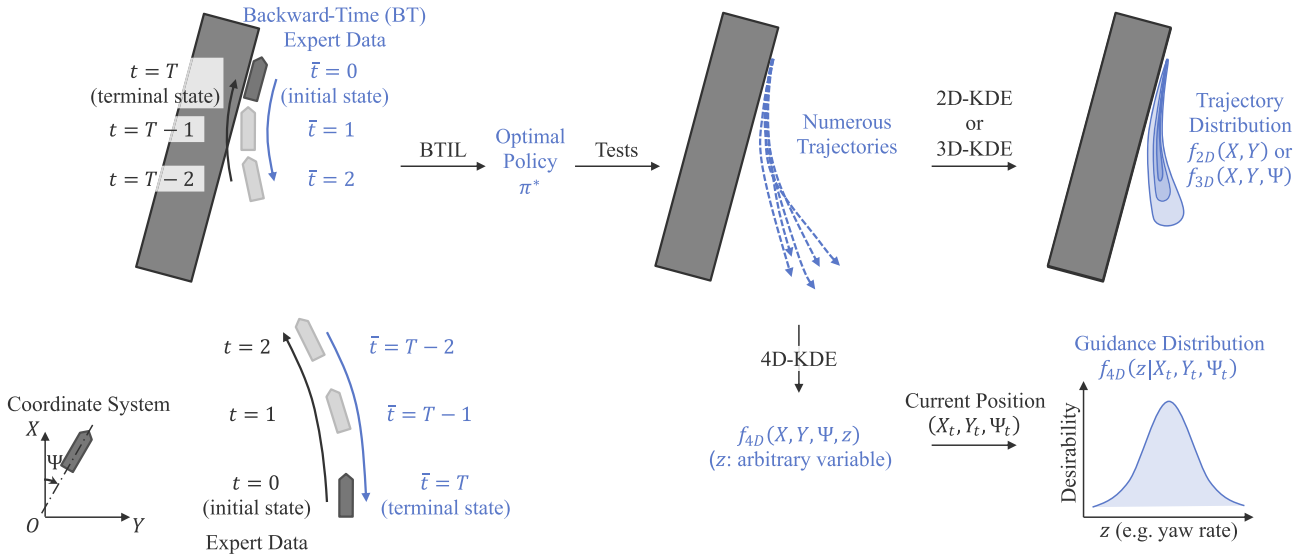


Fig. 1. Overview of the docking assistance method, which consists of BTIL using AIS data, rollouts of the obtained policy, computation of trajectory distributions by 2D and 3D-KDE, and guidance distributions by 4D-KDE.

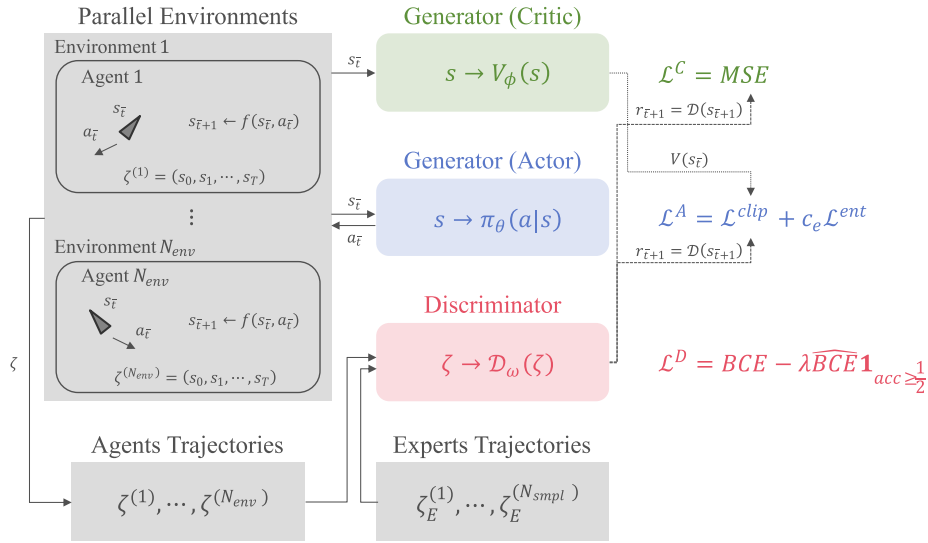


Fig. 2. Computation flow of the BTIL for docking trajectory planning.

Algorithm 1. Backward-time imitation learning (BTIL)

1. Reverse the time histories of the expert data: $\bar{t} = T - t$.
2. Initialize the weight parameters of discriminator, actor, and critic: ω , θ , and ϕ .
3. while training do
4. Initialize the state, $s_{\bar{t}=0}$, using the terminal state of the expert data, $s_{E, \bar{t}=0}$.
5. while $\bar{t} < T$ do
6. Stochastically determine an action following the current policy: $a_{\bar{t}} \sim \pi_{\theta}(a|s_{\bar{t}})$.
7. Transition the state: $s_{\bar{t}+1} \leftarrow f(s_{\bar{t}}, a_{\bar{t}})$.
8. Collect the history of the states: $\zeta = (s_0, \dots, s_{\bar{t}+1})$.
9. Update the weight of discriminator to minimize the loss, $\mathcal{L}^D(\omega)$.
10. Calculate the discriminator's rewards, $(r_1, \dots, r_{\bar{t}+1})$.
11. Update the weights of actor and critic to minimize the losses, $\mathcal{L}^A(\theta)$ and $\mathcal{L}^C(\phi)$.

3. Methods

3.1. Overview of docking assistance method

Fig. 1 shows an overview of the docking assistance method presented in this study. First, the docking trajectories obtained from AIS data were reversed over time. We then conducted BTIL, in which the agent learned the experts' policy while returning in time from the terminal states of the expert data. After sufficient BTIL training, many trajectories were generated using the optimized stochastic policy π^* . Finally, the trajectory and guidance distributions were derived by applying KDE to the generated trajectories.

3.2. Trajectory planning by BTIL

Fig. 2 describes the computational flow of BTIL for docking trajectory planning. The trajectory planner consists of three NNs: generator (actor), generator (critic), and discriminator. For fast and stable computation, we employed distributed learning, in which multiple

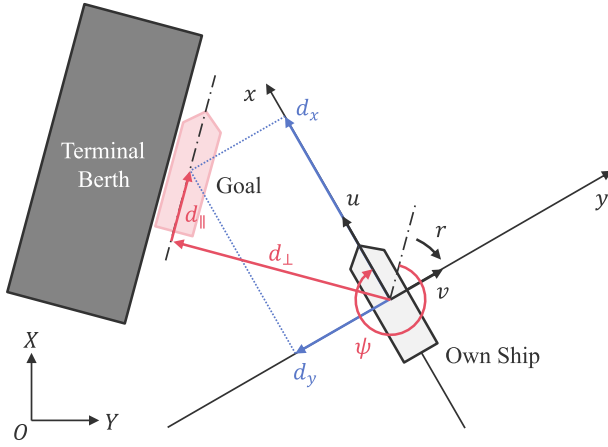


Fig. 3. Schematic of the state space corresponding to the own ship and destination.

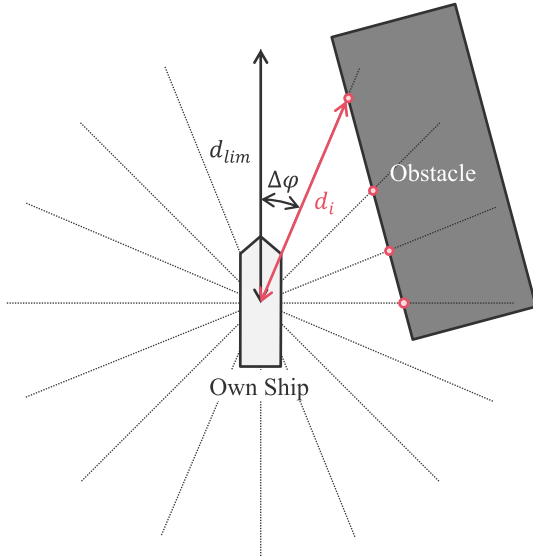


Fig. 4. Schematic of the state space corresponding to obstacle detection.

agents took actions and received transitioned states in parallel N_{env} environments. In many IL, the generator uses the logarithm of the discriminator's output as a reward. Then, the reward always takes negative values; as a result, an agent is likely to learn to terminate an episode as early as possible. To avoid this issue, this study defined the reward as $r_{t+1} = \mathcal{D}(s_{t+1})$, which always takes a positive value.

Expert trajectories were first prepared using AIS data, which included timestamps, longitudes, latitudes, speeds over the ground, courses over the ground, and heading angles of the target ship. As the timestamp intervals were irregular, we interpolated the time history of each data using a quadratic spline curve and created data for every second. Next, the drift angles, surge velocities u , and sway velocities v were calculated from the courses over the ground and the heading angles. Angular velocities (yaw rates) were also calculated by the linear difference as $r_t = (\Psi_{t+1} - \Psi_t)/dt$. The surge, sway, and yaw accelerations were computed as $\dot{u}_t = (u_{t+1} - u_t)/dt$, $\dot{v}_t = (v_{t+1} - v_t)/dt$, and $\dot{r}_t = (r_{t+1} - r_t)/dt$. To mitigate the noise in the angular accelerations while keeping its influence as minimal as possible, we smoothed them by taking the moving average with a period $M = 2$, with all time histories reversed.

The state spaces for both the agent and expert in the trajectory planning environment are defined as follows. The state was divided into

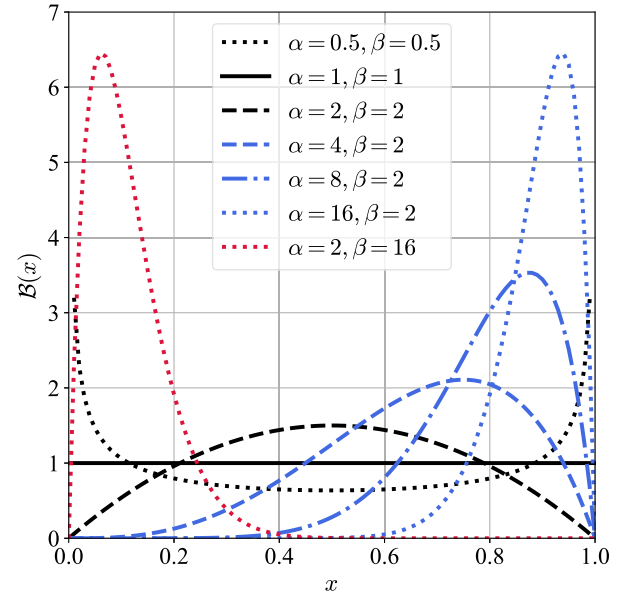


Fig. 5. Examples of probability density functions of beta distributions with different α and β .

two categories: one corresponding to the own ship and destination, S_{base} , and the other corresponding to the detection of obstacles, S_{detect} . As illustrated in Fig. 3, S_{base} was composed of the longitudinal component of the relative position of the destination from the own ship d_x , transverse component d_y , parallel component to the terminal berth $d_{||}$, vertical component $d_{\perp}$, deviation of heading angle of the own ship from the berth ψ , surge velocity component of the own ship u , sway component v , and yaw rate r . For practical use, all the components were normalized within the range $[-1, 1]$,

$$S_{base} = [d'_x, d'_y, d'_{||}, d'_{\perp}, \psi', u', v', r'] \quad (5)$$

where the prime symbol denotes that the variable is normalized. Detection lines are effective for detecting obstacles, as demonstrated by Shen et al. (2019). In this study, S_{detect} was defined using the detection lines radiating from the ship, as shown in Fig. 4.

$$S_{detect} = [d'_1, \dots, d'_n] \quad (6)$$

$$\text{where } d'_i = \begin{cases} \frac{d_i}{d_{lim}} & \text{if obstacle exists} \\ 1 & \text{if obstacle does not exist} \end{cases}$$

Here, d_i denotes the distance of an obstacle from the own ship on detection line i . d_{lim} is the length of the detection line, that is, the maximum detectable range.

The action space was defined as the surge, sway, and yaw angular accelerations of the ship.

$$A = [a^1, a^2, a^3] = [\dot{u}, \dot{v}, \dot{r}] \quad (7)$$

The ship maneuvering model or control signals were not formulated in the action space because this study focused on trajectory planning rather than motion control. Instead, our planner implicitly considered the ship's maneuvering motion by constraining the policy based on AIS data. Following the previous work by Chou et al. (2017), we applied the following beta policy:

Table 1
Principal hyperparameters of the three NNs.

	Generator (Actor)	Generator (Critic)	Discriminator
Hidden Layer	3	3	2
Hidden Nodes	64	64	64
Activation	tanh	tanh	ELU
Learning Rate	1×10^{-5}	1×10^{-5}	5×10^{-5}

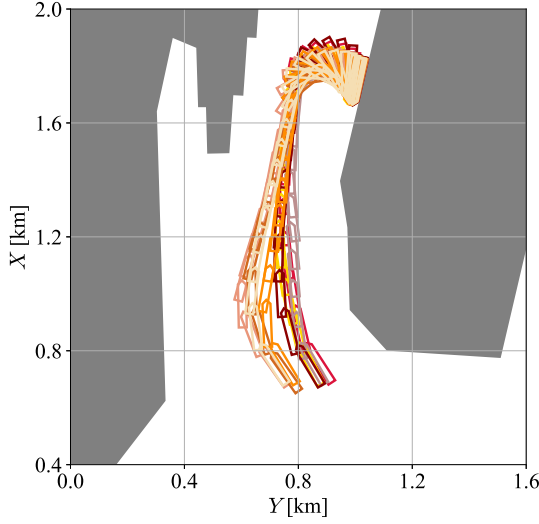


Fig. 6. Docking trajectories of Housho-Maru on eight days: December 5th, 9th, and 13th, January 19th and 27th, and February 10th, 18th, and 26th. The position of the ship is plotted every minute.

$$\pi(a^j|s) = \begin{cases} \mathcal{B}\left(\frac{a^1 + |\dot{u}|_{\max}}{2|\dot{u}|_{\max}}; \alpha^1, \beta^1\right), \\ \mathcal{B}\left(\frac{a^2 + |\dot{v}|_{\max}}{2|\dot{v}|_{\max}}; \alpha^2, \beta^2\right), \\ \mathcal{B}\left(\frac{a^3 + |\dot{r}|_{\max}}{2|\dot{r}|_{\max}}; \alpha^3, \beta^3\right) \end{cases} \quad (8)$$

where $\mathcal{B}(x; \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$

This policy is assumed to follow a beta distribution of the first kind, $\mathcal{B}(x; \alpha, \beta)$. Fig. 5 demonstrates examples of the probability density functions of beta distribution with different parameters, α and β . Since beta distribution has a support $x \in [0, 1]$, it can easily constrain the range of actions. Also, it can represent various distributions using only two parameters. In this study, during both the learning and testing phases, actions were stochastically selected by sampling from the distribution, although they could be deterministically selected using the mean value of the distribution if needed. The maximum values of each action, $|\dot{u}|_{\max}$, $|\dot{v}|_{\max}$, and $|\dot{r}|_{\max}$, were set based on the AIS data, as discussed in Section 4.1.

Table 1 lists the principal hyperparameters of the three NNs used in the proposed trajectory planner. Each NN was optimized using Adam (Kingma and Ba, 2014). For the PPO as an actor, the clipping value and entropy coefficient were set as $\epsilon = 0.2$ and $c_e = 1 \times 10^{-4}$. For regularizing the discriminator, the dropout technique (Srivastava et al., 2014) with the rate $\rho = 0.8$ was applied before the first and after the second hidden layers. For implementing TRAIL, we set the interval of the invariant set as $T_{\mathcal{I}} = 20$ and the coefficient as $\lambda = 1.0$. This study assumed a berthing problem as an MDP with the time step size $dt = 5$ [s] and discount rate $\gamma = 0.99$. The time horizon, i.e., the maximum number

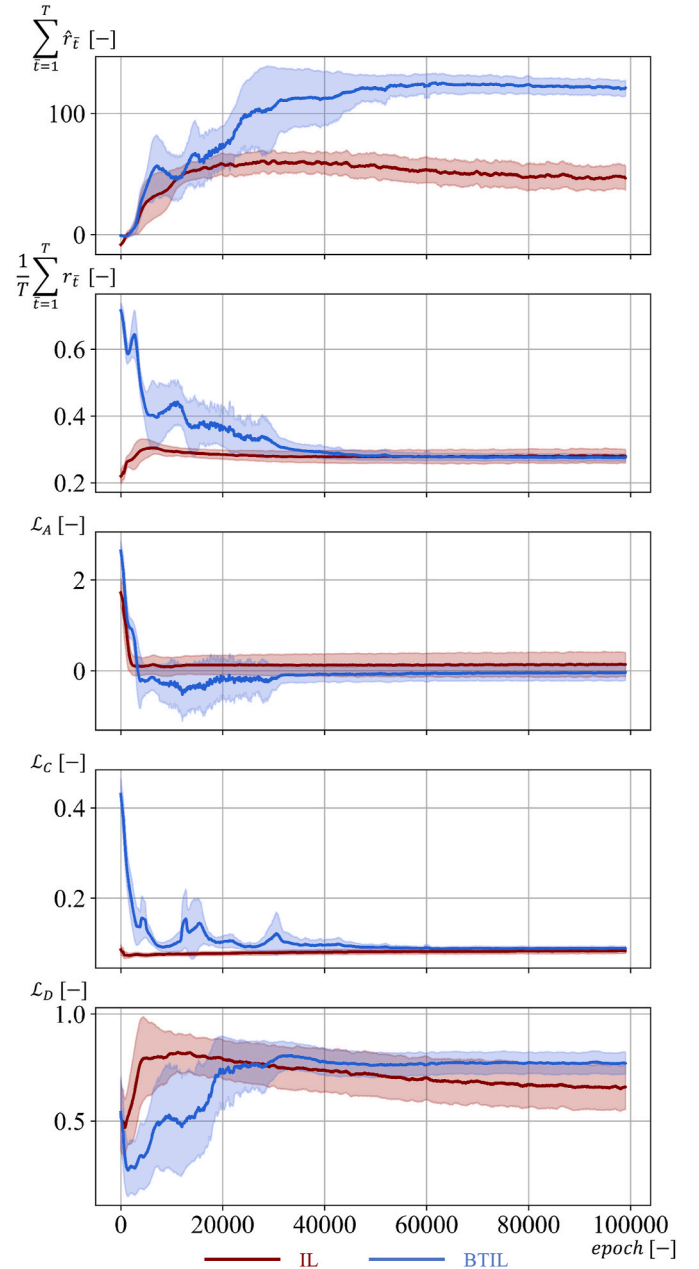


Fig. 7. Learning curves of IL and BTIL. From top to bottom: cumulative reward from the environment, average reward from the discriminator, and loss of the actor, critic, and discriminator. The solid lines and shaded areas indicate the means and standard errors in three training trials with different random seeds. These values were smoothed by moving average with the period $M = 1000$. Note that the reward from the environment, $\hat{r}_t$, was provided for reference purposes only and did not play any role in the learning.

of steps per episode, was set at $T = 300$. The episode was terminated when a collision occurred with a quay. The own ship had 16 detection lines at intervals of $\Delta\varphi = 22.5$ [°], with their length set as $d_{lim} = 1$ [km]. For distributed learning, the agent was trained in a $N_{env} = 16$ parallel environment.

3.3. Trajectory and guidance distributions

The proposed method introduced three kinds of KDE: spatial KDE with respect to $X - Y$ coordinates (2D-KDE), KDE with respect to $X - Y - \Psi$ coordinates (3D-KDE), and KDE with respect to $X - Y - \Psi$ coordinates and an arbitrary variable (4D-KDE). By applying 2D-KDE to

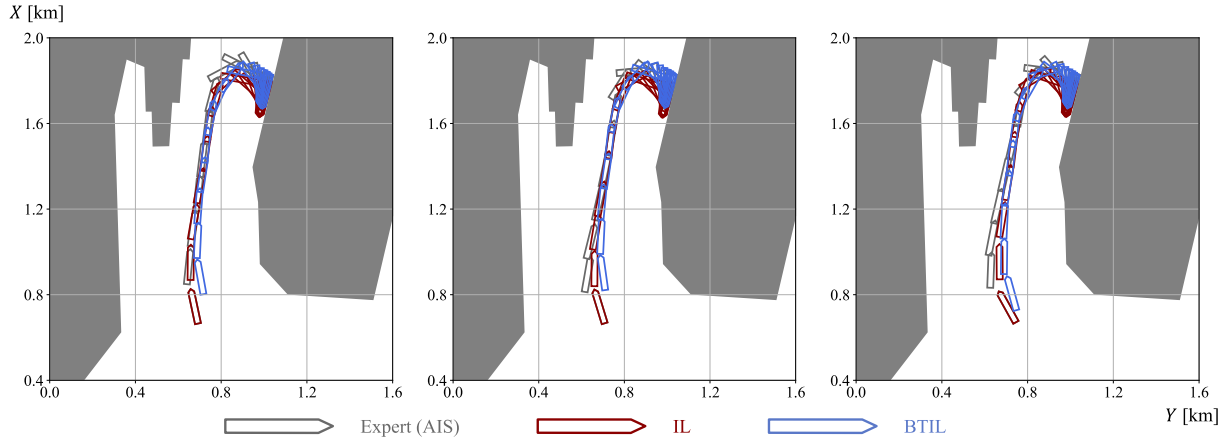


Fig. 8. Comparisons among the actual expert trajectories obtained from the AIS data, forward-time trajectory plans generated by IL, and backward-time trajectory plans generated by BTIL. The expert trajectories were obtained from the AIS data of Toyofuji-Marui on January 15th and 24th and February 12th (from left to right). The tests on IL began from the initial conditions of the expert ($s_{E,t=0}$), whereas the tests on BTIL began from the terminal conditions of the expert ($s_{E,\bar{t}=0}$).

the generated trajectories consisting of N time steps in total, a trajectory distribution can be obtained as follows:

$$f_{2D}(X, Y) = \frac{1}{Nh^2} \sum_{i=1}^N K\left(\frac{X - X_i}{h}\right) K\left(\frac{Y - Y_i}{h}\right) \quad (9)$$

where $K(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

where h is a parameter called the bandwidth that determines the

smoothness of the distribution. Similarly, a trajectory distribution considering the heading angles can be obtained using the following 3D-KDE:

$$f_{3D}(X, Y, \Psi) = \frac{1}{Nh^3} \sum_{i=1}^N K\left(\frac{X - X_i}{h}\right) K\left(\frac{Y - Y_i}{h}\right) K\left(\frac{\Psi - \Psi_i}{h}\right) \quad (10)$$

In addition, a guidance distribution for an arbitrary variable z can be acquired using 4D-KDE.

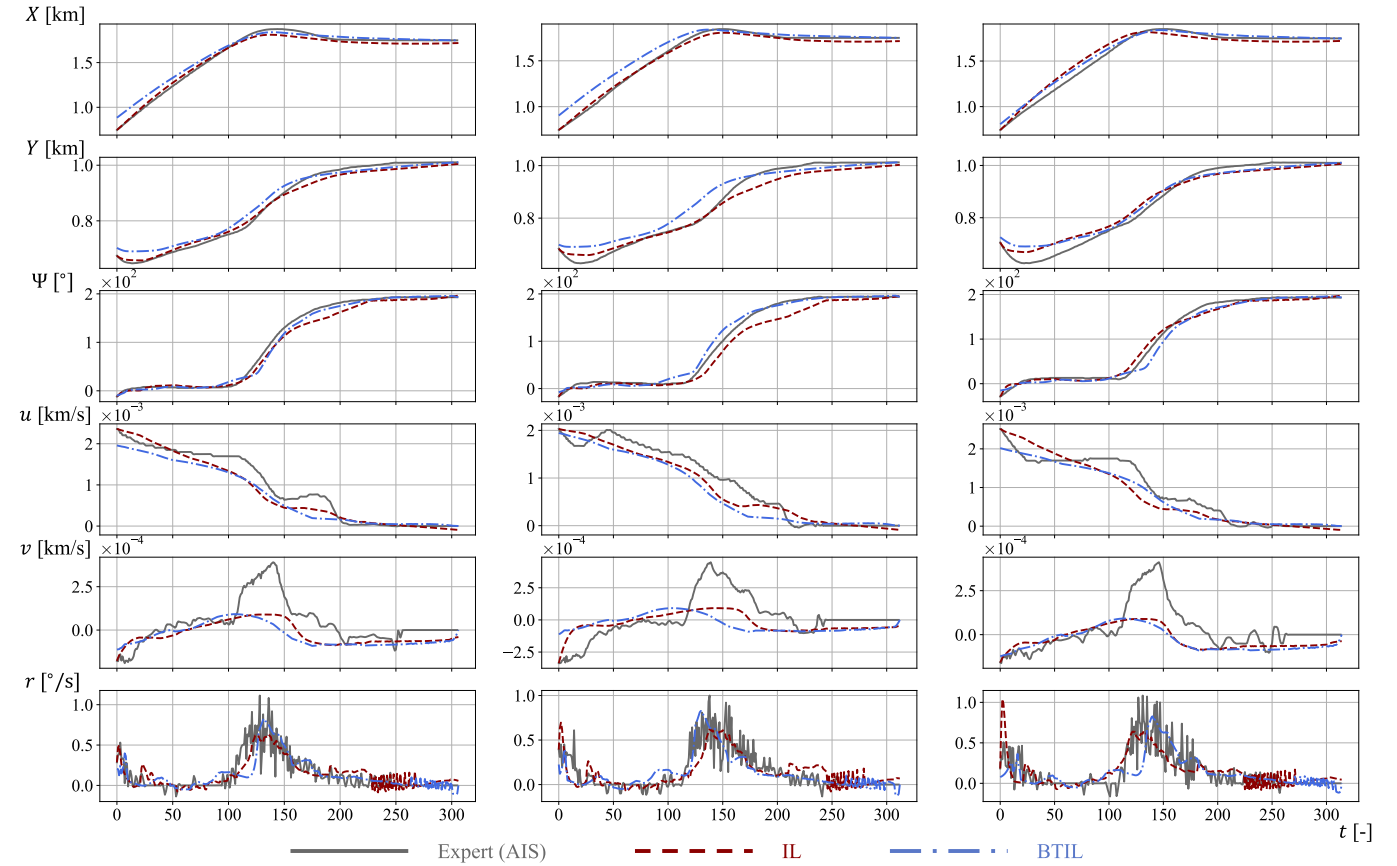


Fig. 9. Time histories of the latitudinal positions, longitudinal positions, heading angles, surge velocities, sway velocities, and yaw rates obtained from the AIS data, IL, and BTIL. From left to right: target scenario of the docking of Toyofuji-Marui on January 15th and 24th and February 12th, respectively. Note that the reversed time histories generated by BTIL were re-reversed in time.

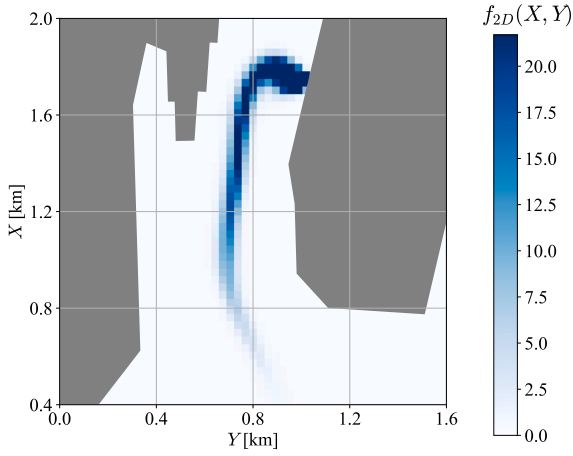


Fig. 10. Trajectory distribution calculated by 2D-KDE.

$$f_{4D}(X, Y, \Psi, z) = \frac{1}{Nh^4} \sum_{i=1}^N K\left(\frac{X - X_i}{h}\right) K\left(\frac{Y - Y_i}{h}\right) K\left(\frac{\Psi - \Psi_i}{h}\right) K\left(\frac{z - z_i}{h}\right) \quad (11)$$

Then, the guidance distribution corresponding to the current position (X_t, Y_t, Ψ_t) can be attained as

$$f_{4D}(z|X_t, Y_t, \Psi_t) \quad (12)$$

Table 2

Conditions for judging successful berthing (Ishibashi, 2020). Each variable is visually described in Fig. 13.

Key Variable	Definition	Condition
Berthing angle	Angle between the berth and the ship	$ \psi \leq 6^\circ$
Entry angle	Angle between the normal line to the berth and the course over ground	$ \psi_U \leq 12^\circ$
Berthing speed	Ship speed that is measured from the berth	$U \cos \psi_U \leq 0.05$ [m/s]
Eccentric distance	Distance between the center line of the berth and the center of gravity of the ship	$ d_{ } \leq 15$ [m]

For the target variable z , we chose the surge, sway, and angular velocities u , v , and r in this study.

4. Results

4.1. Expert data

The docking assistance method was verified using actual trajectories at the Shinmoji Port in Fukuoka Prefecture, Japan, where many vessels travel. The geometry of the virtual port for the simulation experiments was created using longitudes and latitudes available on Google Maps. The coordinate origin was located at $130^\circ 59'$ E and $33^\circ 52'$ N.

The AIS data obtained in December 2021 and January and February 2022 were used as the expert trajectories. A vehicle carrier—Housho-Marun, 165.0 [m] in overall length and 27.6 [m] in overall width and equipped with bow and stern thrusters—was selected as the training

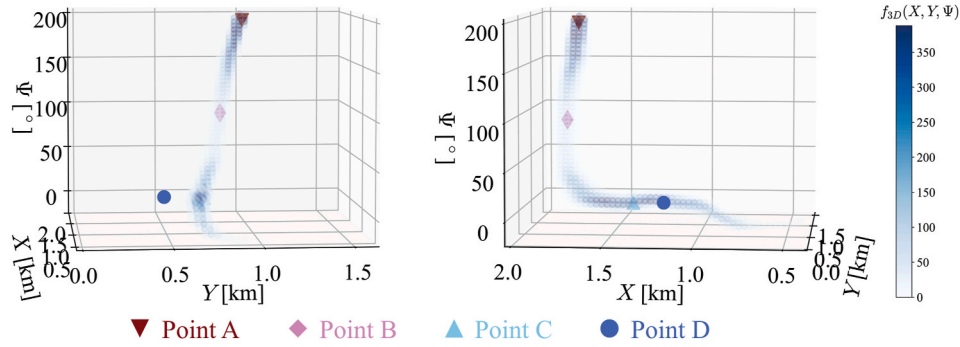


Fig. 11. Trajectory distribution obtained from 3D-KDE in two different views.

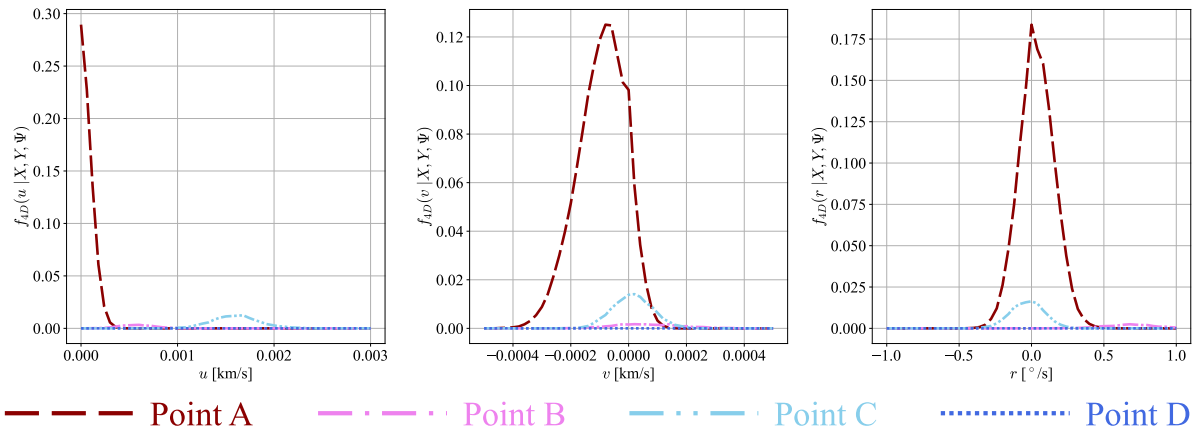


Fig. 12. Guidance distributions at the points A, B, C, and D.

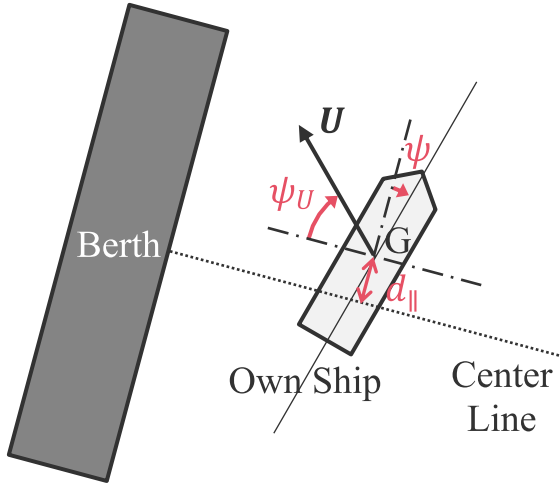


Fig. 13. Definition of key variables for judging successful berthing.

dataset. Fig. 6 shows the actual docking trajectories on December 5th, 9th, and 13th, January 19th and 27th, and February 10th, 18th, and 26th. Based on the actual accelerations, the upper and lower bounds of the action spaces were determined as $|\dot{u}|_{\max} = 1.6 \times 10^{-5} [\text{km}/\text{s}^2]$, $|\dot{v}|_{\max} = 1.4 \times 10^{-5} [\text{km}/\text{s}^2]$, and $|\dot{r}|_{\max} = 0.18 [^\circ/\text{s}^2]$. By substituting these values into Eq. (8), we implicitly constrained the accelerations within feasible ranges, specifically for the target ship. In this verification, heading angles were defined within the range of $[-90^\circ, 270^\circ]$ to prevent discontinuity.

Looking at the AIS data, it was confirmed in advance that the target ship was not supported by tugboats. Wind data at Kukokita-machi were collected from the Japan Meteorological Agency database. The maximum wind speed was 4.9 [m/s] during docking over the eight days; however, no significant differences were observed in the trajectories owing to wind conditions. According to the reanalysis database named JCOPE-t, provided by Forecast Ocean Plus, the maximum current speed encountered by the target ship was 0.23 [m/s]. Hence, the effects of wind and current on trajectory planning were assumed to be negligible in this verification, except for the discussion part (Section 5).

4.2. Trajectory planning with IL and BTIL

Fig. 7 shows the learning curves of IL and BTIL, from top to bottom, the cumulative rewards from the environment, average rewards from the discriminator, and losses of the actor, critic, and discriminator. Note that the reward from the environment, $\hat{r}_t$, was provided for reference purposes only and did not play any role in learning (the reward was set to increase as the ship approached the destination and rapidly decreased if a collision occurred; however, to avoid confusion, this paper does not provide detailed definitions). Although BTIL did not demonstrate a clear advantage over IL in the learning process, its convergence across the three trials was confirmed.

The trajectory planner based on BTIL was tested in comparison with IL using three actual trajectories of Toyofuji-Marui. Toyofuji-Marui, a sister ship of Housho-Marui, has the same hull form and destination. Figs. 8 and 9 compare the expert trajectories obtained from the AIS data and the planned trajectories computed by IL and BTIL. The imitation results computed using IL and BTIL were consistent with the expert data. Even for the yaw rate, which involved considerable noise in the expert data, both the IL and BTIL achieved an appropriate estimation. While the terminal positions and velocities calculated by the IL differed slightly from those of the expert, those calculated by BTIL were constrained to the expert. Although BTIL cannot match the initial condition to the current situation, this will be addressed by the integration with KDE, as presented in the next section.

4.3. Trajectory distributions

After sufficient training of BTIL, 10,000 trajectories were generated by testing the obtained policy from the destination: $X_0 = 1.7474 [\text{km}]$, $Y_0 = 1.0080 [\text{km}]$, and $\Psi_0 = 193.00 [^\circ]$. Applying 2D-KDE to the backward-time histories of the generated trajectories, we acquired a stochastic trajectory distribution, as shown in Fig. 10. We can grasp the area through which the ship should pass for successful docking but without information on the heading angles.

Subsequently, 3D-KDE was introduced into the generated trajectories. Fig. 11 shows the trajectory distribution, including the heading angles. Compared to the 2D-KDE distribution, the 3D-KDE distribution seems more informative, although it may be less intuitive, i.e., more difficult to understand at a glance. Sampling points A, B, C, and D were used to verify the guidance distributions in Section 4.4.

4.4. Guidance distributions

In addition to the trajectory distributions obtained by 2D and 3D-KDE, we computed 4D-KDE to obtain guidance distributions regarding velocities and yaw rates at arbitrary locations. Fig. 12 shows the distributions at the four sampling points plotted in Fig. 10: $(X_A, Y_A, \Psi_A) = (1.7474, 1.0080, 193.00)$, $(X_B, Y_B, \Psi_B) = (1.8000, 0.8800, 90.00)$, $(X_C, Y_C, \Psi_C) = (1.4000, 0.7500, 4.00)$, and $(X_D, Y_D, \Psi_D) = (1.2000, 0.5000, 10.00)$. At the destination (point A), the guidance distributions peaked at approximately zero, consistent with the desired terminal state. The distribution of the sway velocity v exhibited a peak at slight speed, probably because of the coarse resolution when calculating the probability distributions. In this study, $X - Y - \Psi$ space was discretized into a square grid with intervals of 32 [m] to calculate the distributions numerically; hence, point A also had the size. Accordingly, the expert data had a slight sway speed, which led to a shift in the guidance distributions. During a 180-degree turn (point B), small peaks appeared at approximately $u = 0.6 [\text{m}/\text{s}]$ and $r = 0.7 [^\circ/\text{s}]$. The peak values at point B were much smaller than at point A because the combination of X , Y , and Ψ was enormous. At point C, the distribution suggests that the ship should move at approximately 1–2 [m/s], allowing minimal drift and rotation. At point D, outside the trajectory distribution, the guidance distributions had almost zero values, indicating that the proposed method can also identify whether a ship is within the range of human-like trajectories.

To verify the guidance distributions, we made an agent continue taking actions that maximize each distribution. In this validation, the initial positions and heading angles were comprehensively set as follows: X_0 was divided into 50 segments within the range of $[0.4, 2.0] \text{ km}$, Y_0 within the range of $[0.0, 1.6] \text{ km}$, and Ψ_0 within the range of $[-20, 200]$ degree, resulting in $50^3 = 125,000$ scenarios. The initial velocities were determined to maximize the guidance distributions.

$$u_0 = \operatorname{argmax}_u f_{4D}(u|X_0, Y_0, \Psi_0) \quad (13)$$

$$v_0 = \operatorname{argmax}_v f_{4D}(v|X_0, Y_0, \Psi_0) \quad (14)$$

$$r_0 = \operatorname{argmax}_r f_{4D}(r|X_0, Y_0, \Psi_0) \quad (15)$$

From each initial position, the agent was forced to take the following accelerations at each time step:

$$\dot{u}_{t+1} = \frac{\operatorname{argmax}_u f_{4D}(u|X_t, Y_t, \Psi_t) - u_t}{dt} \quad (16)$$

$$\dot{v}_{t+1} = \frac{\operatorname{argmax}_v f_{4D}(v|X_t, Y_t, \Psi_t) - v_t}{dt} \quad (17)$$

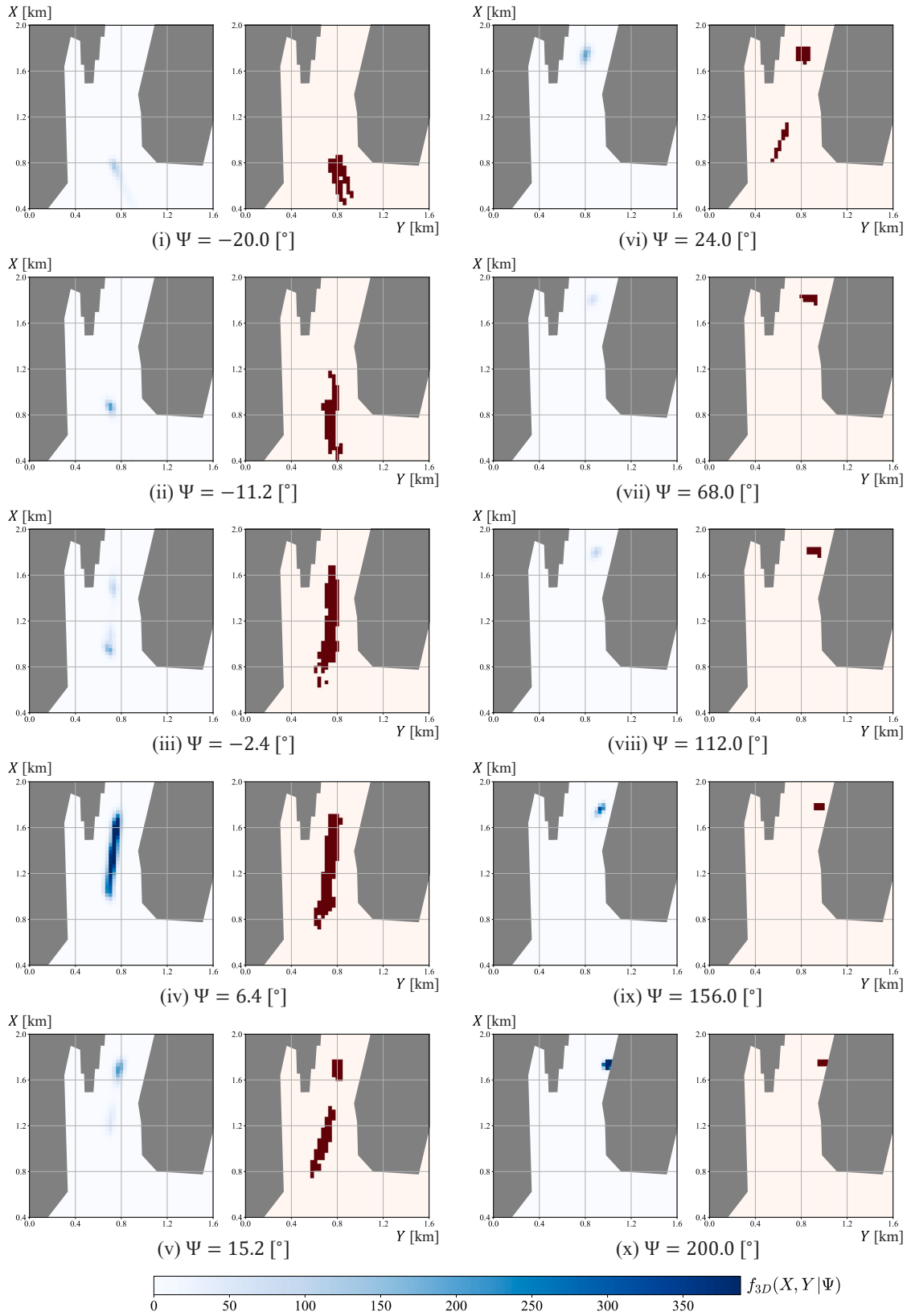


Fig. 14. Success maps in comparison with the trajectory distributions on the planes of (i) $\Psi = -20.0$ [°], (ii) $\Psi = -11.2$ [°], (iii) $\Psi = -2.4$ [°], (iv) $\Psi = 6.4$ [°], (v) $\Psi = 15.2$ [°], (vi) $\Psi = 24.0$ [°], (vii) $\Psi = 68.0$ [°], (viii) $\Psi = 112.0$ [°], (ix) $\Psi = 156.0$ [°], and (x) $\Psi = 200.0$ [°]. The dark red areas represent successful docking domains, from which the ship successfully docked to the berth by trusting the guidance distributions. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

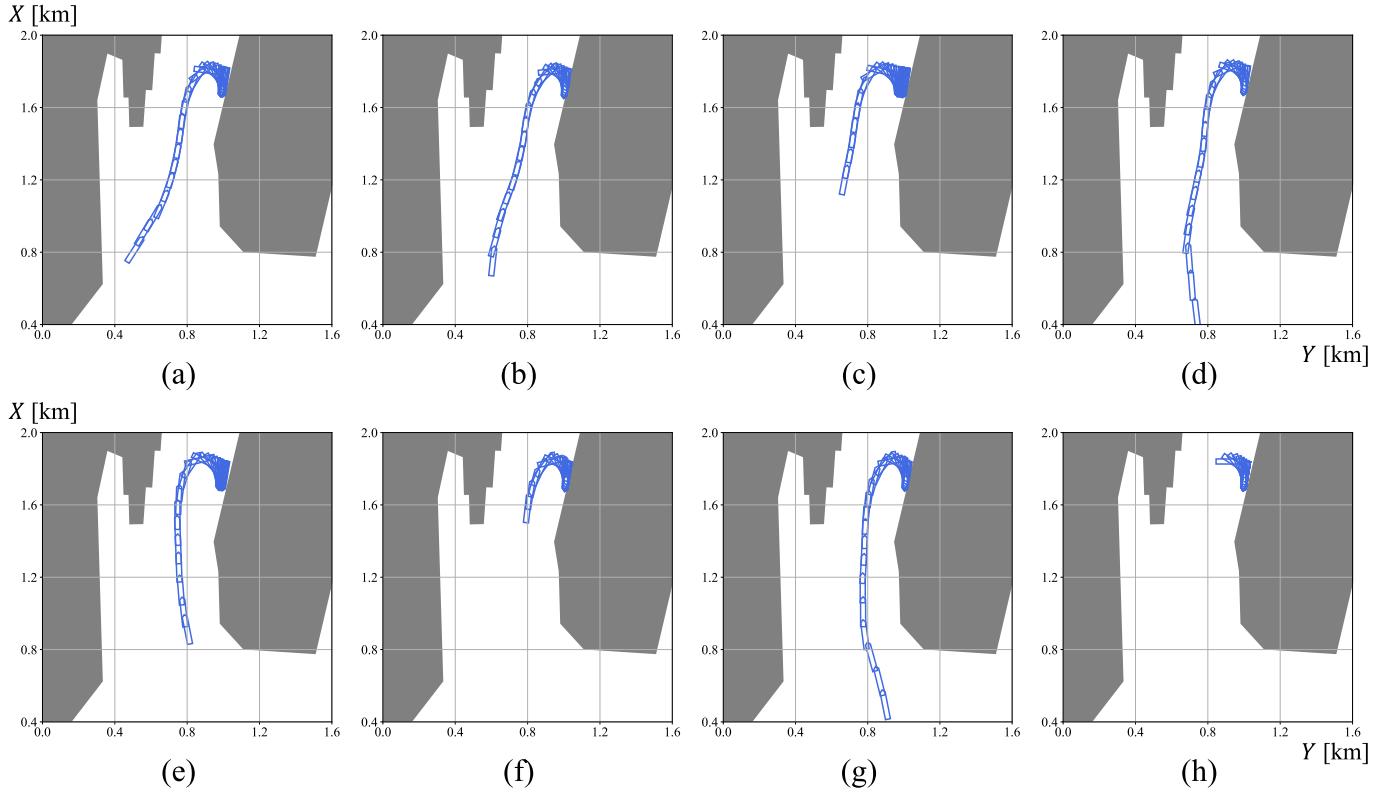


Fig. 15. Examples of tested trajectories obtained by taking the accelerations calculated from Eqs. (16)–(18), from the random initial positions shown in Table 2. The position of the ship is plotted every minute.

Table 3
Examples of random initial positions.

Case	X_0 [km]	Y_0 [km]	Ψ_0 [°]
(a)	0.784	0.480	28.4
(b)	0.720	0.576	2.0
(c)	1.168	0.640	6.4
(d)	0.432	0.704	− 11.2
(e)	0.880	0.768	− 15.6
(f)	1.552	0.768	2.0
(g)	0.464	0.864	− 15.6
(h)	1.808	0.896	85.6

$$\dot{r}_{t+1} = \frac{\arg\max_r f_{4D}(r|X_t, Y_t, \Psi_t) - r_t}{dt} \quad (18)$$

The acquired trajectories were then judged whether successful or not. Here, a docking trajectory was judged as “successful” if it simultaneously satisfied the four conditions shown in Table 2 (Ishibashi, 2020). In this study, the center of gravity of the ship was assumed to be at the midship point.

Fig. 14 shows the success maps compared with the trajectory distributions. The dark red areas represent the successful docking domains, from which the ship successfully docked to the berth by trusting the guidance distributions. Comparing each trajectory distribution (shown on the left-hand side) and the success map (shown on the right-hand side), the success map appears to cover the trajectory distribution on every $X - Y$ plane, indicating that if the ship is within a domain wherein the trajectory distribution is above a certain threshold, it can dock properly by following the distribution from that point onward. Docking was successful under 462 initial conditions. In contrast, when we directly applied KDE to the expert data without BTIL for comparison, the guidance distributions resulted in only 274 successful cases. The comparison indicates that training and sampling with BTIL enhanced the

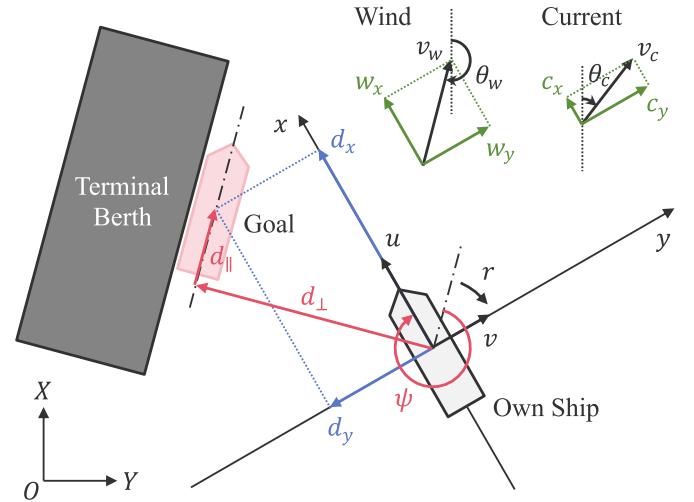


Fig. 16. Schematic of the enhanced state space corresponding to the own ship, destination, and disturbances.

generalization performance of the proposed method. Fig. 15 shows examples of the tested trajectories from the random initial positions listed in Table 3. The results suggest that appropriate docking plans can be obtained by trusting and following guidance distributions.

5. Discussions

In the preceding sections, we have assumed that environmental disturbances are negligible. This is because the wind or current did not show significant influence on the docking trajectories, as noted in Section 4.1. However, this is not always the case; especially during bad

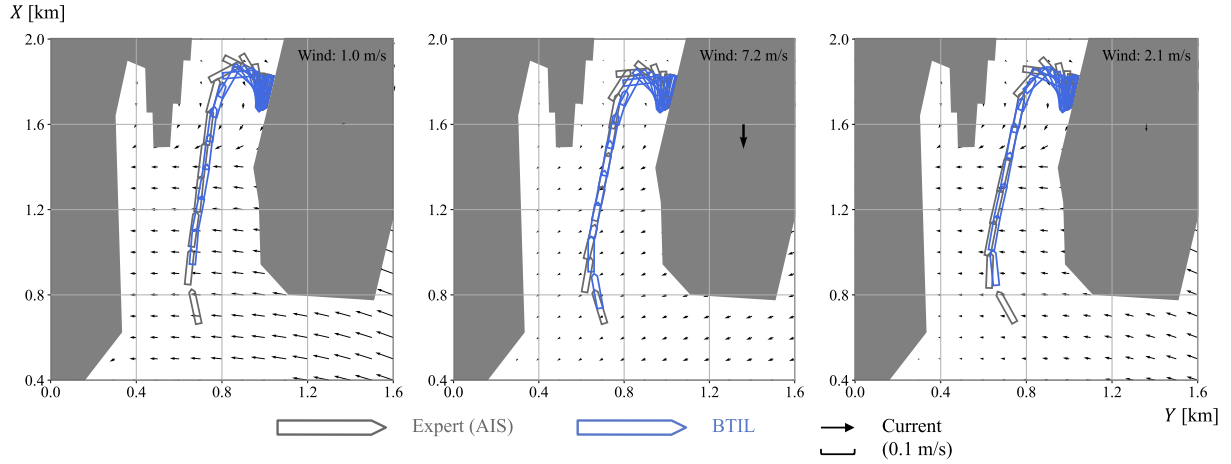


Fig. 17. The backward-time trajectory plans generated by the enhanced BTIL, which are compared with the expert trajectories obtained from the AIS data of Toyofuji-Maru on January 15th and 24th and February 12th (from left to right).

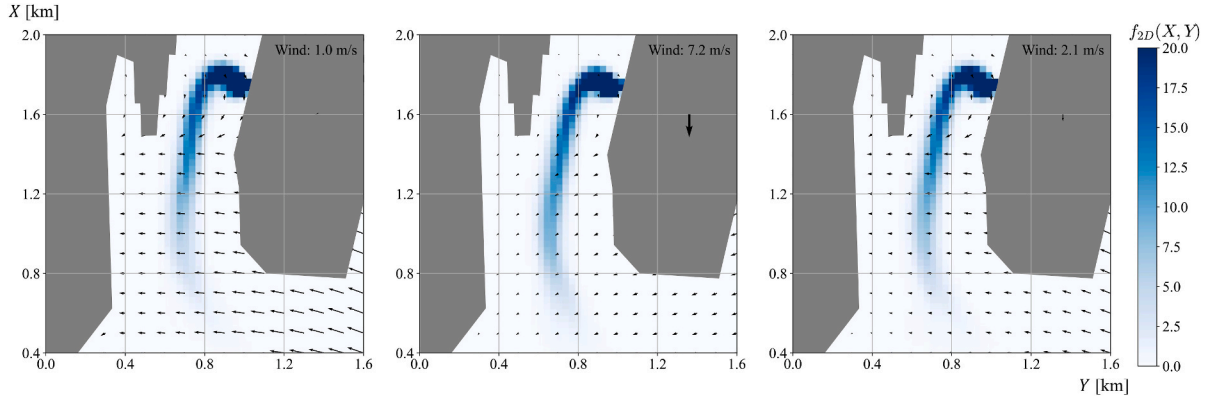


Fig. 18. Trajectory distributions under the same wind and current conditions that Toyofuji-Maru likely encountered on January 15th and 24th and February 12th (from left to right).

weather, captains are likely to consider such disturbances when planning a docking trajectory. To prepare for practical implementation, we confirmed that the proposed method can incorporate wind and current velocities. Fig. 16 shows the schematic of the enhanced state space: w_x and w_y represent the wind velocities in surge and sway directions, and c_x and c_y represent the current velocities in surge and sway directions, respectively. These four components were divided by the representative speeds and added to the S_{base} as follows:

$$S_{base} = [\hat{d}_x, \hat{d}_y, \hat{d}_{\parallel}, \hat{d}_{\perp}, \hat{\psi}, \hat{u}, \hat{v}, \hat{r}, \hat{w}_x, \hat{w}_y, \hat{c}_x, \hat{c}_y] \quad (19)$$

The representative wind and current speeds were set as 5 [m/s] and 0.25 [m/s], which were approximate maximum values during the docking of Housho-Maruo on the eight days (December 5th, 9th, and 13th, January 19th and 27th, and February 10th, 18th, and 26th).

The wind data, obtained from the database of [Japan Meteorological Agency](#), were measured at Kukokita-machi, located at 131°21' E and 33°50.7' N. The average wind speeds and directions per hour were available, and we used the values at the time closest to that of docking. The current data were obtained from the reanalysis database named JCOPE-t ([Forecast Ocean Plus](#)). Since the data were available at 1-h and 1 km intervals, we linearly interpolated them to 100 m intervals.

Fig. 17 shows the backward-time trajectory plans generated by the enhanced BTIL, which are compared with the expert trajectories obtained from the AIS data of Toyofuji-Maruo on January 15th and 24th and February 12th. Although, as mentioned above, we did not find

significant impact of disturbances on the trajectories, the enhanced BTIL replicated the expert trajectories with equal or better accuracy than the results without considering disturbances (Fig. 8). Fig. 18 depicts the trajectory distributions under the same wind and current conditions that Toyofuji-Maruo likely encountered on January 15th and 24th and February 12th. We confirmed that the 2D trajectory distributions could be successfully generated.

6. Conclusions

This study proposed a novel docking assistance method that combined BTIL and KDE. The proposed method was verified using actual docking trajectories at the Shinmoji Port in Fukuoka Prefecture, Japan. First, both IL and BTIL were conducted using the AIS data of a vehicle carrier, Housho-Maruo. Testing with its sister ship, Toyofuji-Maruo, the imitation results computed by the IL and BTIL showed good agreement. While the terminal positions and velocities calculated by the IL differed slightly from those of the expert, the BTIL's results were constrained to the expert. Subsequently, 10,000 trajectories were generated using the obtained stochastic policy. The docking assistance method provided trajectory and guidance distributions after applying 2D, 3D, and 4D-KDE to the generated trajectories. The two-dimensional trajectory distributions suggested the area through which a ship should pass for successful docking. The three-dimensional distributions were more informative because they included the heading angles. The guidance distributions derived from the 4D-KDE indicated the velocities the ship should maintain. To verify the guidance distributions, we placed an agent at

comprehensive initial positions and heading angles and made it consistently perform actions that maximize each distribution. Consequently, the success maps covered the trajectory distributions on every $X - Y$ plane; hence, if the ship is within a domain wherein the trajectory distribution is above a certain threshold, it can dock properly by following the distribution from that point onward. In this test, docking was successful for 462 initial conditions; however, applying KDE directly to expert data (without BTIL) resulted in only 274 successful cases. This implies that training and sampling using BTIL enhanced the generalization performance of the proposed method. The discussion section suggested that the docking assistance method has the potential to provide trajectory distributions adaptively while considering external disturbances. The aforementioned verification was also conducted using the AIS data of a car ferry named Hibiki (presented in Appendix). All the results showed outcomes similar to those of Housho-Mar, demonstrating the applicability of the proposed method to other vessels, berths, and docking motions.

The proposed method exhibited four advantages over previous ones. First, the IL framework enabled the generation of captain-like docking trajectories without setting any objective functions. Finding objective functions that reproduce the sophisticated decision-making of human experts is challenging; however, our research addressed this issue. Second, preparing expert data was relatively easy, requiring only AIS and port geometry data. By generalizing expert trajectories through BTIL, only a minimal amount of data was required. Third, the proposed method provided distributions rather than a single route. As the distributions are numerically differentiable, it may be easy to integrate them with other control systems, such as MPC. Moreover, the proposed method can be used to determine the permissible range of autonomous navigation systems. Specifically, setting a threshold for the distribution may allow human operations if the distribution falls below the threshold. Finally, this method offered advantages in terms of computational speed. Once pre-trained, it can generate distributions in real time. Therefore, the proposed method can provide appropriate guidance that corresponds to the changing ship conditions and environmental disturbances.

In future studies, we plan to address the following three challenges:

- 1) Validation on disturbance effects: Although this study suggested the potential of the proposed method to treat disturbances, the validation was insufficient due to the data limitation. In particular, the wind and current data may have differed from the actual conditions experienced by the target ship. By using onboard data, we will investigate the impact of disturbances on docking trajectories and provide docking trajectory distributions that account for the disturbances.
- 2) Investigation of distribution thresholds: Currently, the threshold values for the trajectory and guidance distributions, i.e., the minimum values required to ensure proper operation, are unknown. For

practical applications, determining the thresholds that guarantee safe navigation through experiments and further investigations is necessary.

- 3) Experimental validation: In this paper, the effectiveness of the proposed method is only shown through simulations. For further development of this study, we aim to demonstrate the proposed method can achieve appropriate docking in combination with other control technologies at real-world experiments.

CRediT authorship contribution statement

Takefumi Higaki: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Hirofumi Hashimoto:** Writing – review & editing, Resources, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT (GPT-4o) in order to check grammar and spelling. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Takefumi Higaki reports financial support was provided by Japan Society for the Promotion of Science. Hirofumi Hashimoto reports financial support was provided by Japan Society for the Promotion of Science. Takefumi Higaki reports financial support was provided by Fundamental Research Developing Association for Shipbuilding and Offshore. Hirofumi Hashimoto reports financial support was provided by ClassNK. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Application to Different Ship and Berth

To investigate the applicability of the proposed method, we applied it to a car ferry named Hibiki—195.0 [m] in overall length and 29.6 [m] in overall width and equipped with bow and stern thrusters. During the target six months—December 2021 and January, February, June, July, and August 2022—59 trajectories were available, with some shown in Fig. A.1. Based on the actual accelerations, the upper and lower bounds of the action space were determined as $|\dot{u}|_{\max} = 2.2 \times 10^{-5} [\text{m}/\text{s}^2]$, $|\dot{v}|_{\max} = 1.4 \times 10^{-5} [\text{m}/\text{s}^2]$, and $|\dot{r}|_{\max} = 0.1 [^\circ/\text{s}^2]$. The time horizon was set as $T = 200$.

Figs. A.2, A.3, A.4, A.5, and A.6 show the trajectory distributions by 2D and 3D-KDE, guidance distributions, success maps, and tested trajectories. All results showed similar outcomes to those of the Housho-Mar, demonstrating the applicability of the proposed method to other vessels and berths. While direct KDE without BTIL led to 674 successful cases, KDE with BTIL was successful in 4437 initial conditions. Hence, training and sampling with BTIL enhanced the generalization performance of the proposed method.

Table A.1
Examples of random initial positions.

Case	X_0 [km]	Y_0 [km]	Ψ_0 [°]
(a)	0.752	0.640	9.4
(b)	0.464	0.672	4.0
(c)	1.360	0.704	1.6
(d)	1.104	0.736	− 14.0
(e)	0.368	0.800	0.4
(f)	0.560	0.928	− 20.6
(g)	0.368	1.024	− 18.2
(h)	0.432	1.248	− 15.2

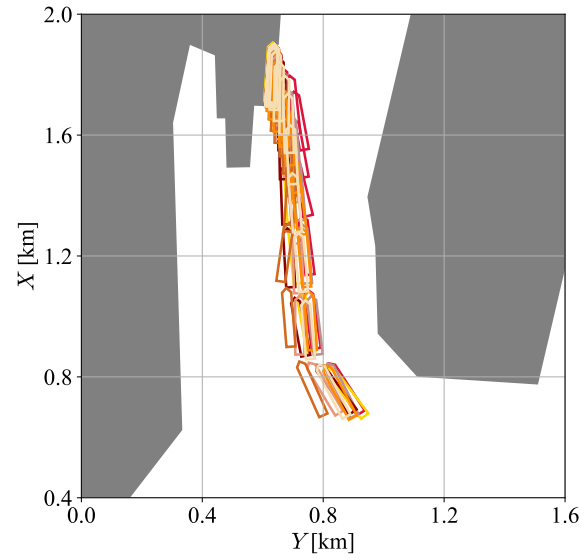


Fig. A.1. Several docking trajectories of Hibiki. The position of the ship is plotted every minute.

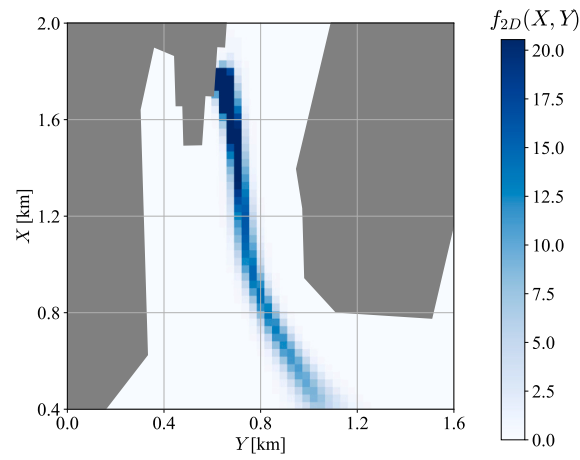


Fig. A.2. Trajectory distribution calculated by 2D-KDE.

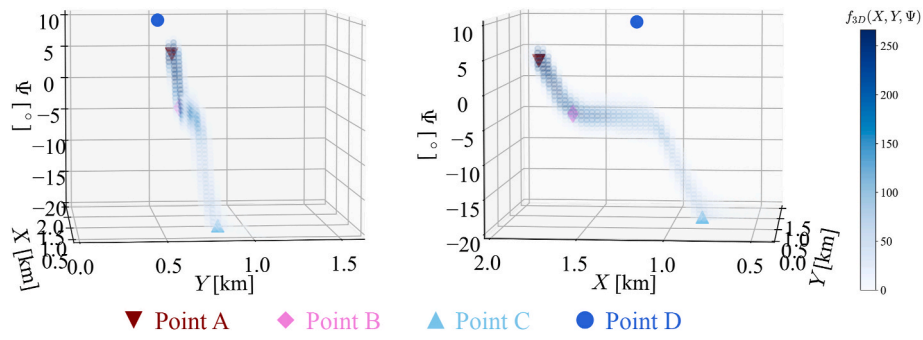


Fig. A.3. Trajectory distribution obtained from 3D-KDE in two different views.

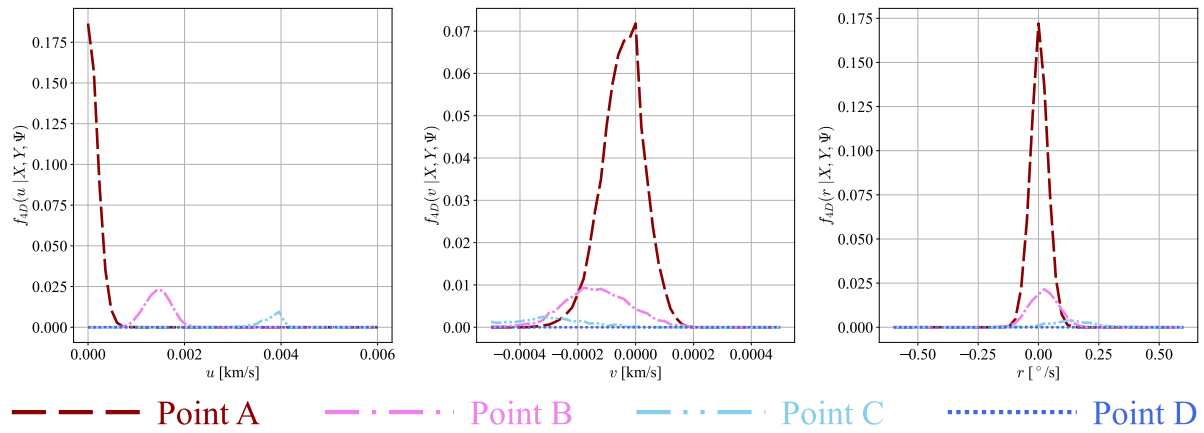


Fig. A.4. Guidance distributions at the points A, B, C, and D.

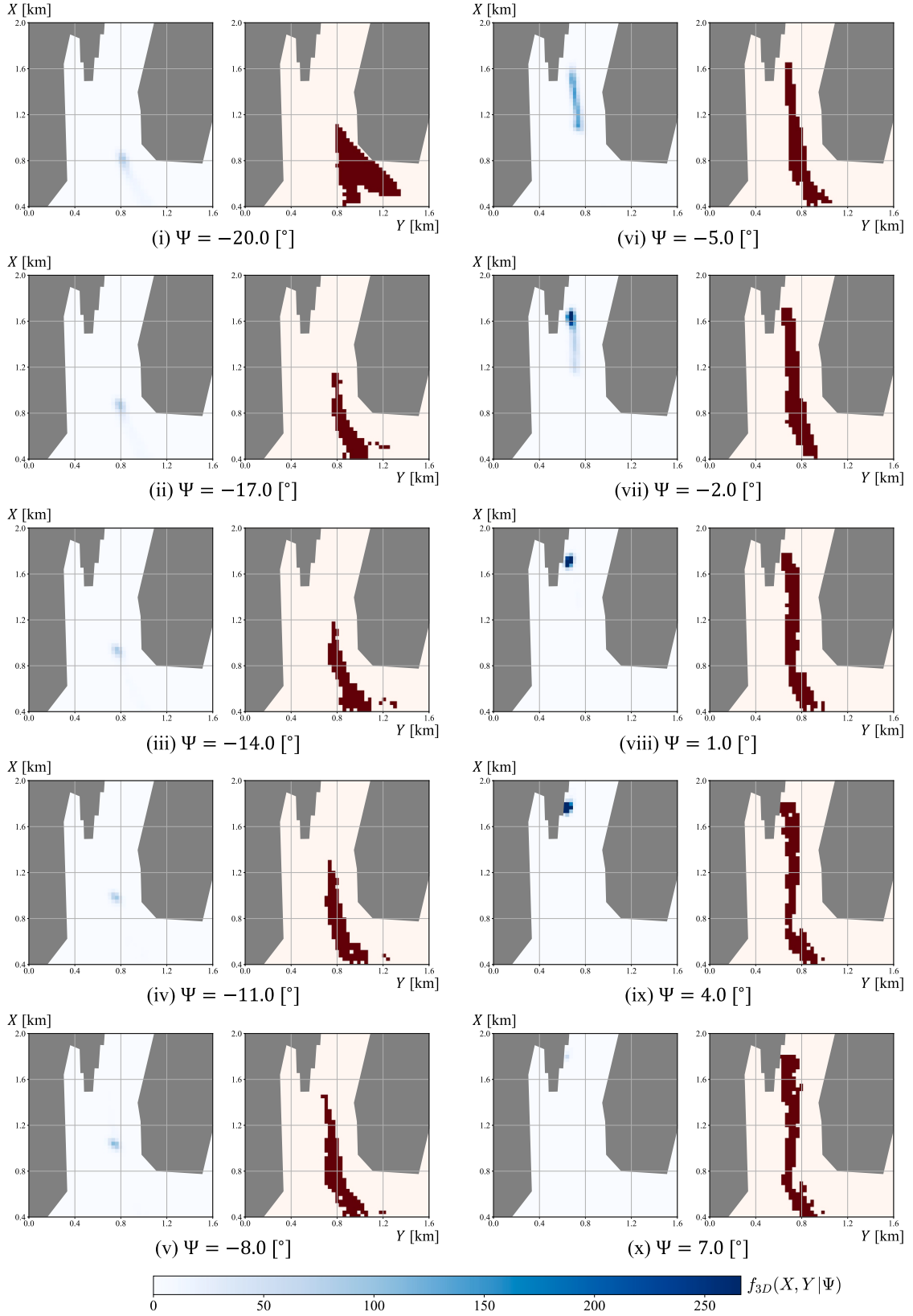


Fig. A.5. Success maps in comparison with the trajectory distributions on the planes of (i) $\Psi = -20.0$ [°], (ii) $\Psi = -17.0$ [°], (iii) $\Psi = -14.0$ [°], (iv) $\Psi = -11.0$ [°], (v) $\Psi = -8.0$ [°], (vi) $\Psi = -5.0$ [°], (vii) $\Psi = -2.0$ [°], (viii) $\Psi = 1.0$ [°], (ix) $\Psi = 4.0$ [°], and (x) $\Psi = 7.0$ [°]. The dark red areas represent successful docking domains, from which the ship successfully docked to the berth by trusting the guidance distributions.

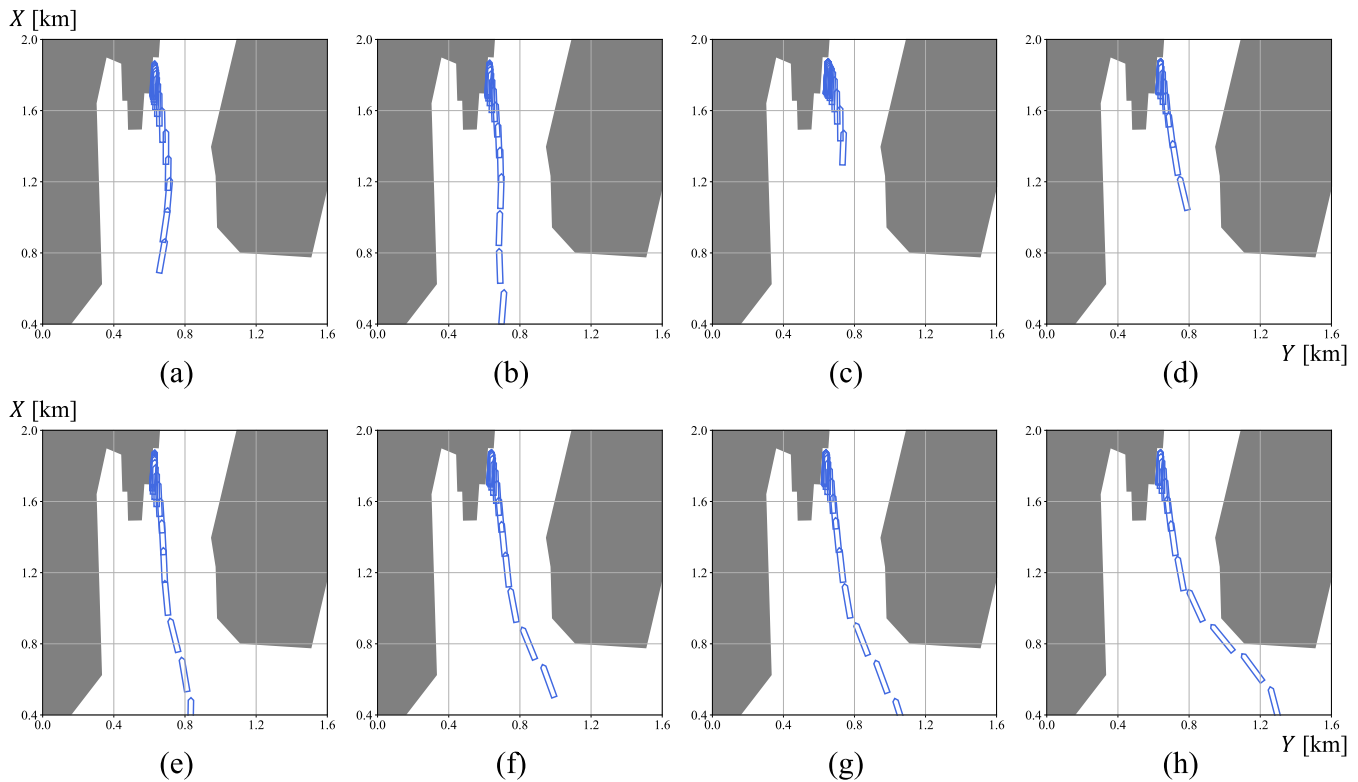


Fig. A.6. Examples of tested trajectories obtained by considering the accelerations calculated from Eqs. (16)–(18), based on the random initial positions shown in Table A.1. The position of the ship is plotted every minute.

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